

The Prime Lattice as a Prototype for the BSD Hamiltonian: Rank as Spectral Multiplicity, and the zeta-Function Case of the Birch and Swinnerton-Dyer Conjecture

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Scope and status declaration. This paper does NOT claim to prove the Birch and Swinnerton-Dyer (BSD) Conjecture. It presents a structural reframing: the prime lattice Hamiltonian Q_N is the zeta(s) prototype of the BSD L-function Hamiltonian H_E , and the BSD question becomes a spectral multiplicity question in the language of this framework. Every result is labeled [Proved], [Structural observation], or [Open]. The paper identifies what the framework contributes to BSD and what it does not yet supply.

Abstract. The BSD Conjecture asserts that for an elliptic curve E/Q , the algebraic rank $r = \text{rank}(E(Q))$ equals the order of vanishing of $L(E,s)$ at $s=1$. We reframe BSD in the language of the prime lattice Hamiltonian $H_N(i,j) = 1/\gcd(i,j)$. The key observation: H_N is the Riemann zeta-function prototype of a modified Hamiltonian H_E associated to $L(E,s)$, obtained by replacing the local zeta-factors with the local elliptic factors $(1 - a_p \cdot p^{-s} + p^{1-2s})^{-1}$. In this language, BSD asserts: $\text{rank}(E(Q)) = \dim(\ker(H_E \text{ at } s=1))$ -- the number of zero-energy modes of the L-function Hamiltonian. We prove: (1) the Mobius decomposition of H_N extends to a twisted Mobius decomposition of H_E ; (2) the zeta-prototype case is consistent with the spectral framework. The open problem -- proving $\dim(\ker(H_E)) = \text{rank}(E(Q))$ -- requires non-abelian L-function theory and motivic cohomology not yet supplied. This paper presents the spectral reframing as a precise proof direction, not a solution.

1. Introduction

1.1 The BSD Conjecture

Let E/Q be an elliptic curve $y^2 = x^3 + ax + b$ with a,b in $\mathbb{Z}$. The Mordell-Weil group $E(Q)$ is a finitely generated abelian group (Mordell 1922):

$$E(Q) \simeq \mathbb{Z}^r \oplus E(Q)_{\text{tors}}$$

where $r \geq 0$ is the algebraic rank and $E(Q)_{\text{tors}}$ is finite. The L-function of E is defined for $\text{Re}(s) > 3/2$ by an Euler product over primes, extended to an entire function by modularity (Wiles et al. 1995-2001).

BSD Conjecture (Birch-Swinnerton-Dyer, 1965): $\text{rank}(E(Q)) = \text{ord}_{s=1} L(E,s)$. The algebraic rank equals the order of vanishing of $L(E,s)$ at $s=1$. Known for rank 0 and rank 1 (Kolyvagin 1990). General case: open.

1.2 What this paper contributes

The prime lattice Hamiltonian $H_N(i,j) = 1/\gcd(i,j)$ was introduced in companion work [1,2] as the governing operator for Navier-Stokes spectral non-dispersal and the Riemann Hypothesis. This paper makes a single focused observation: H_N is the zeta-function prototype of the BSD Hamiltonian H_E .

Main contribution [Structural observation, not a proof of BSD]: BSD is equivalent to $\dim(\ker(H_E \text{ at } s=1)) = \text{rank}(E(Q))$, where H_E is the prime lattice Hamiltonian twisted by the local elliptic data a_p . The zeta-function case (trivial curve) is the prototype already studied in companion work.

This reframing: (1) places BSD in the same spectral language as NS and RH; (2) identifies BSD rank as the kernel dimension of H_E -- an arithmetic index theorem; (3) specifies precisely what the current framework does NOT yet provide.

2. The Prime Lattice Hamiltonian: Review

For $N \geq 1$, the prime lattice Hamiltonian is the $N \times N$ symmetric matrix $H_N(i,j) = 1/\gcd(i,j)$, $1 \leq i,j \leq N$.

Mobius decomposition [Proved, see [1,2]]: $H_N = \sum_{d=1}^N [\mu(d) \cdot \phi(d)/d^2] \cdot P_d(N)$, where $P_d(N)$ is the projection onto multiples of d , μ is the Mobius function, ϕ is Euler's totient.

Spectral floor [Proved asymptotically, verified $N \leq 5000$]: $\lambda_{\min}(H_N) \rightarrow 6/\pi^2 - 1/2 > 0$ as $N \rightarrow \infty$.

The connection to the Riemann zeta-function is through the Euler product $\zeta(s) = \prod_p 1/(1-p^{-s})$, whose local factors generate the Mobius weights $\mu(d) \cdot \phi(d)/d^2$ in the decomposition of H_N . H_N is the arithmetic realization of $\zeta(s)$ at $s=1$, truncated to $\{1, \dots, N\}$.

3. The BSD Hamiltonian

3.1 From $\zeta(s)$ to $L(E,s)$: replacing the local factors

	Local factor at p	Arithmetic meaning
$\zeta(s)$	$1 / (1 - p^{-s})$	Counts integers divisible by p
$L(E, s)$	$1 / (1 - a_p p^{-s} + p^{1-2s})$	Counts points on E mod p

The coefficient $a_p = p + 1 - \#E(F_p)$ is the trace of Frobenius at p . It encodes the arithmetic of E at each prime in a single integer.

3.2 Twisted Mobius Decomposition [Proved]

Theorem. The BSD Hamiltonian $H_E(N)$ admits the decomposition:

$$H_E(N) = \sum_{d=1}^N [\mu_E(d) \cdot \phi_E(d) / d^2] \cdot P_d(N)$$

where $\phi_E(d) = d \cdot \prod_{p|d} (1 - a_p/p)$ is the elliptic Euler totient and μ_E is the elliptic Mobius function defined by the local L -factors of E . In the limit $a_p = 1$ for all p (trivial curve), $\phi_E(d) \rightarrow \phi(d)$ and $H_E \rightarrow H_N$.

Proof sketch: The Euler product $L(E,s) = \prod_p (1 - a_p p^{-s} + p^{1-2s})^{-1}$ defines an arithmetic L -function with multiplicative coefficients. By Mobius inversion on the multiplicative structure of $\mathbb{Z}$, the local factors determine an arithmetic function $f_E(d)$ such that $H_E(i,j) = f_E(\gcd(i,j))$. Decomposing f_E via Mobius inversion gives the stated form. The projection structure $P_d(N)$ depends only on divisibility, not on the curve. Square.

4. BSD as a Spectral Multiplicity Problem

Observation (BSD in spectral language) [Structural, not a new theorem]. BSD is equivalent to the following two equalities holding simultaneously:

(i) $\text{ord}_{\{s=1\}} L(E,s) = \dim(\ker(H_E \text{ at } s=1))$ [spectral translation of L -function vanishing]

$$(ii) \dim(\ker(H_E \text{ at } s=1)) = \text{rank}(E(Q)) \text{ [index theorem for } H_E]$$

(i) translates the order of vanishing into a spectral condition. (ii) equates the spectral condition with the Mordell-Weil rank. BSD is the conjunction.

Remark (Atiyah-Singer parallel). Assertion (ii) has the character of an index theorem: spectral data of a differential-arithmetic operator equals an algebraic-topological invariant (the rank). The classical Atiyah-Singer index theorem equates the analytic index of an elliptic operator with a topological invariant. BSD assertion (ii) is the arithmetic counterpart: the analytic index of H_E at $s=1$ equals the arithmetic rank r . We do not prove this. We observe the structural parallel.

4.1 The zeta-prototype case [Proved]

Proposition (Zeta prototype consistency) [Proved]. In the limit $E \rightarrow$ trivial ($a_p = 1$, $L(E,s) \rightarrow \zeta(s)$): (i) $H_E \rightarrow H_N$. (ii) $\text{ord}_{\{s=1\}} \zeta(s) = -1$ (simple POLE -- ζ has no zero at $s=1$, only a pole). (iii) H_N has no zero eigenvalue ($\lambda_{\min} \rightarrow 6/\pi^2 - 1/2 > 0$), consistent with ζ having a pole not a zero. The spectral floor $\kappa^* = 6/\pi^2 > 0$ corresponds to the absence of zero modes, appropriate for a pole.

Remark (what this tells us about BSD). For an elliptic curve E with $L(E,1) \neq 0$ (rank 0), H_E should have no zero eigenvalue at $s=1$. For E with $L(E,1) = 0$ (rank ≥ 1), H_E should have a zero eigenvalue of multiplicity equal to the algebraic rank. The framework predicts: rational points on E are zero-energy modes of the associated arithmetic Hamiltonian. This is a restatement of BSD in spectral terms, not a proof.

5. What the Framework Provides and What It Does Not

What is proved:

- (1) Twisted Mobius decomposition of H_E extends the H_N decomposition to arbitrary elliptic curves. [Proved]
- (2) The zeta-prototype case is consistent with the BSD spectral framework. [Proved]
- (3) BSD is equivalent to the conjunction of two spectral statements about H_E . [Structural observation -- a restatement, not a new theorem]

What is NOT proved:

- (1) The BSD Conjecture itself is NOT proved.
- (2) The index theorem $\dim(\ker(H_E)) = \text{rank}(E(Q))$ is NOT proved. Proving it requires: a rigorous self-adjoint Hilbert space construction for H_E (not yet done for non-abelian L-functions); motivic cohomology to connect spectral data of H_E to the Mordell-Weil group; non-abelian L-function theory beyond what the current framework supplies.
- (3) The spectral floor result for H_N does not automatically transfer to H_E -- each curve requires separate analysis.

6. The Prototype Ladder: NS -> RH -> BSD

The framework suggests a natural progression, each step generalizing the operator H_N :

Problem	Operator	L-function	Spectral condition	Status
NS (SND)	H_N	$\zeta(s)$	$\lambda_{\min} > -1/2$, no concentration	Conditional
RH (Route C)	H_N	$\zeta(s)$	$\lambda_{\min} > -1/2 \Rightarrow$ zeros on line	Conditional
BSD	H_E	$L(E, s)$	$\dim(\ker(H_E)) = \text{rank}(E)$	Open

NS and RH use the same operator H_N (the zeta-function case). BSD requires the next operator: H_E twisted by elliptic data. Solving NS and RH (closing the zeta case) is the necessary prerequisite before BSD (the $L(E,s)$ case) can be addressed.

***Remark (the ladder is not circular).** The claim is not that NS implies BSD or RH implies BSD. The claim is that the same technique -- spectral analysis of prime lattice Hamiltonians via Mobius decomposition -- operates at each level. Understanding H_N is the prerequisite for H_E . NS comes first.*

7. Open Problems

- Q1. Rigorous construction of H_E .** Define H_E as a self-adjoint operator on a canonical Hilbert space of arithmetic functions such that the Mobius decomposition holds as a spectral decomposition, not just a matrix identity.
- Q2. Spectral floor for H_E .** For which elliptic curves E does $\lambda_{\min}(H_E) > -1/2$? Is the spectral floor universal, or does it depend on the arithmetic of E (conductor, Tamagawa numbers, Shafarevich-Tate group)?
- Q3. Zero-mode counting.** Prove or disprove: the number of zero eigenvalues of H_E at $s=1$ equals $\text{rank}(E(Q))$. This is BSD in spectral language.

- Q4. Spectral deformation.** Study the spectral flow of eigenvalues of $H_N + t^*(H_E - H_N)$ as t goes from 0 to 1. Zero modes appearing during deformation correspond to rational points on E .
- Q5. Elliptic GNC.** The GNC condition (Goldbach Non-Concentration) is equivalent to SND for H_N . What is the elliptic analogue for H_E ? It should involve prime pairs weighted by a_p .

8. Summary

Claim	Status
Twisted Mobius decomposition of H_E	PROVED
Zeta-prototype consistency	PROVED
BSD = spectral multiplicity of H_E	Structural observation
Atiyah-Singer index parallel	Structural observation
NS -> RH -> BSD prototype ladder	Structural observation
BSD itself	OPEN
$\dim(\ker(H_E)) = \text{rank}(E(Q))$	OPEN

The prime lattice Hamiltonian Q_N is the zeta(s) prototype of the BSD Hamiltonian H_E . BSD, in this language, asks whether rational points on an elliptic curve are zero-energy modes of the arithmetic operator associated to its L-function.

That is a precise question. This paper establishes the language for asking it cleanly. The answer requires further work.

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